

Viet Dung Nguyen "Dynamic Prior Preference Learning for Scalable, Robust"

1:01:59

hello welcome

back now we have the much anticipated
student presentation that your adviser
Alex recently told us will be awesome by
Viet dong win on Dynamic prior
preference learning for scalable robust
deep active inference so thank you V
take it

away hello uh hello hello everyone um my
name is zet I'm from uh the neuro adap
Computing Laboratory um Department in
computer science seest technology um and
I am uh Alex advis uh and today I'll be
talking about Dynamic PR preference
learning to scale Aus active
INF so yeah first of all want to talk
about the motivation for this so
nowadays you know from fure work in
robotics um excels in implementing
robots behavior and planning for example
um creating um we can you know creating
autonomous robots that can actively
explore in the envir
um additionally um acquiring we can also
do a lot of Robotics applications for
example acquiring knowledge and learn
skills continuously um and um this also
envisioned um cognitive and
developmental robotics um so there just
a goal um for um robots robotics
research um that's we can try to build a
robots that can continuously develop
through interactions with the
environment and their learning processes
um and it is also also better um to have

interactions with their uh physical and social world in the manner of human learning and cognitive development so there's a lot of Robotics applications um around um and for example one application can be to build ASA robots and um building um social collaborative robots and stuff like that um so underlying um all these robot work um uh we it's like active inference research where we um often build a generative models um um it's so the development of Robotics actually depends a lot on active framework because um there's a great need for research um that integrates the generative World model into robotics control system um as also deep learning is increasingly used in constructing complex action planning algorithms for solving control tasks um so um we can see that there's a need for scaling um active inference system especially and deep active INF system um to integrate into the real world applications so um research in active INF has led to many improvements um in terms of robustness of um model based agents that applies to both MDP processes um and partially observable MDPs or POMDPs um so an example of an MDP environment should be Mountain car um uh for those who um knows or studies reinforcement learning already and um this Mountain car environment um in MDP we only know POS velocity right um and that is the actual state of the environment um but when we get to PVP for example like robotic

tasks um we as agents see only the image frame and we do not know the under lying states such as position and orientation of the block or um join state of the robots for example right um so basically active inference is basically a form of inference and learning that sticks to balance first of all the um goal orienting objectives or PRI preference um so this signal exploit um the instrumental signal um so that the agents can seek the goal um and try to pursue its goal um and the other thing it needs to balance the epistemic terms or epistemic signal where the foraging uh it's cares about foraging and exploration driving signals um so researcher often actually encounter great difficulty in providing the agent with effective prior preference as well as Computing useful instrumental signal from the agents the priors um especially in the real world applications where um the environment is very complex the robotic control task and um the trajectory or partway um to be desirable um and Tas specific goals um often requires uh a very very long um trajectory of preference um so it is not that ABS practical to provide uh a prior preference distributions especially the um stream or roll out of go images so that is the reason why we need to find a way to also learn to generate the prior preference signal in active inference so on the right here is just our Umar preliminary results um but we'll get to that in a

moment so

um yes so we actually uh given without
uh all of this uh right we need to
um uh we need to predict a u uh drawout
trajectory of a goal um States
dynamically right um it can be a goal
state or goal observations but basically
the goal is um if you look on the figure
on top here um the top row is the real
trajectory where you see the mountain
car right there's a car that it needs to
go backward and to build some kind of a
momentum and then go full to the right
to get to the um to the flag if it is
only going to the right and it can it
does not have the enough momentum to to
go to to the top of the hill so um so in
here uh is the um actual observations
you can see that the trajectory is um a
successful trajectory and the goal here
is to um for each uh observation at each
time step we predict a um immediate
predict predicted PR preference um image
or goal at that single time step so that
we have um the instrumental signal or
idea of uh what we are trying to learn
and then um from that we learn the
actions planning uh on uh from the PRI
preference
signal so in this talk we are we
specifically analyze this issue um the
issue of um uh prior preference learning
um in the current literature um and
discuss potential resolutions for
scaling the instrumental signal
inherence to active influence by
introducing the um contrasted recurrent
State prior preference or

crsb um this is the model um learning framework that um we also have um done in the past so uh this methodology frames active inference agents in term of progressively constructing and adapt adapting um a prior preference um at each time step so this also facilitate the Dynamics emission of a useful or dense very dense um instrumental signal uh at each time step so we also um study how our prior preference adaptation scheme um when operating jointly with the Laten space driven epistemic foraging terms uh offers active inflece agent um a very nice U flexibility in challenging problem con context um such as um spars to reward um environments um and also composed of uh continuous observation and action space right example um for example robotic task um they only we often like do not know the spars um do not know the dense reward um functions in the actual real environments and we we only know if the robot has risked the goal or not right um and also all these environments has you know varying goal States so um we can also see that providing the uh uh or manually craft the uh distributions of goal is very not very um practical so uh yeah so here here's the outline of this talk on the introduction I've had just covered um so we so first of all I'll just assume that's uh everyone is new to active inference and I'm just going to briefly cover active inference backgrounds and deep active inference um and also cover a little bit

about um prior PRI preference learning
and problems um
uh of all those prior preference
learning framework and then I'm just
going now go through the actual
methodology which is the dynamic Prime
preference learning um and then first of
all is the world model um next is the
self-revision mechanism and then next is
the um learning Prime preference models
and then finally um putting all them
together um to make an agent um and then
um I'll show the results and then
discussion

okay so first of all um I want to um
cover a little bit about active
inference um so just a brief overview
about how active inference work and what
is the underlying theories behind it um
so um it is a ma mathematical framework
that describes how intelligent systems
are human interact with the environment
um so it describe it also describe the
environment as the generative process um
and the intelligent systems um are the
agent or um us as generative model um so
active INF is built on top of the free
energy principles um this means that we
as agent or intelligent system
continuously observe the environment and
forming our and correcting our belief or
our own representation of the
world um so the generative process um
defines a real world environment here
right um this is treated as the data
Generation Um process where the
information patterns are sensed by the
agent so you can see that's the um

hidden State underlying hidden States
 denoted as here u it's generate a sense
 of u some kind of u a observations u
 denoted o and this is also u sensed by
 u our in intelligence
 system so on the right here is the
 generative model u is u the actual
 agent that we are trying to build on the
 intelligent system right u the
 intelligence system also has an internal
 State u which is trying to u estimate
 the generative process u States u so
 this is called u the prior or the
 internal belief prior to the
 observations of the environment at a
 particular time stamp so this is denoted
 as the random variable s or state and P
 of s here u is the distribution over
 the random variable s u so this make
 the agent's generative model u the
 generative model represents the agent's
 approximation of this generative process
 u in this world u by continuously
 seeing the observations or making the
 observation o u and modified or
 updating u its belief so we actually
 call the belief u s after u being
 updated after u making the observation
 or the posterior because it is the
 posterior after u making the
 observation
 right so in order to learn the model and
 estimates the posterior we first need to
 in introduce these terms u first of all
 is the prior over State u which is
 denoted as $P(s)$ which is also the
 distribution over state right u $P(o)$
 here is the model evidence u it is the

distribution over the current
observations u_m denoting how well is our
model at predicting the real data u_m and
in order to compute this easily this is
often written as the sum of probability
uh
of observation given every single
state and then next one is p of O given
 s this is the likelihoods u_m and the
probability uh this is the probability
of an observation given a state so this
is a function it gives a probability
given an input States and then the the
last one is the posterior functions uh p
as given o which is u_m the posterior
function that Maps the observation to
the states u_m so uh normally in
traditional active inference u_m the
peror function would be uh the the
transpose of the likelihood Matrix for
example u_m and deep learning literature
 u_m it is often called a encoder and the
likelihood here is a
decoder so how do we estimate the postor
now u_m we have to find
belief that maximize the modal evidence
right because u_m at the current time
points uh there there is only one single
observation o here and it is real right
so P of o p of that observation u_m it's
going to be uh one right so it should be
the probability of
one u_m and then u_m we use the state to
estimate u_m the observations and then it
has to be the uh the probability of one
two right so uh that arrives us at the
conclusion that we have to u_m maximize
the model evidence

PFO so maximizing P of O is also
 minimizing equal to minimizing minus $\log P$
 of O right because um it is uh
 minimizing the surprise um
 so in the beginning we say that we
 continuously update our belief after
 making an an observations of the
 environment um we call the belief after
 making an observation a posterior um
 this is Q of s
 right um so to compute posterior we want
 to put in a um we just want to um call
 this you know any kind of um
 distributions um over States um then for
 example we can sample it we can
 randomize it right um and then we try to
 gradually trick this distribution so
 that it's used the highest modal
 evidence or it's used the lowest
 surprise so how do we choose U the
 actual uh posterior um so that it bus
 the lowest surprise right so in minus $\log P$
 of O here is the actual um uh
 minimizing surprise that objective that
 we are wanting to to minimize right so
 we can uh analyze it like uh we can uh
 do it with a base theorem this not Bas
 theorem but base formula or theories
 yeah um and then
 uh to evaluate the surprise minus $\log P$
 of O we need to sum out um the hidden
 variable s from The Joint distribution
 POS um
 but well this summations there's a
 problem right because that we do not
 know like all the hidden States all the
 hidden variables in the environments um
 it is hidden from us right so it is

normally very very hard to uh to sum up
all these um State values so uh if we
cannot sum it up then we can try to
approximate it

so first we introduce a posterior um in
into this uh sum right we multiply both
the denominator and the numerator by the
posterior um so this is the belief that
we are trying to tweak right so um in
here we're trying to um took this
posterior so that um we can um minimize
this

term so this gives us a r sum where s uh
the ratio $\frac{P(s)}{Q(s)}$ and um over the the
distributions um Q of s is weed by the Q
of s this is like weed sum
right so according to the inequality of
convexity or gen inequality says that's
the convex function of a weighted means
is always smaller um than the weed means
of the same convex function um so you
can see

the convex function here is minus log
of something right

so minus log of something is just a
hyper uh is just a parabol something
like here um and uh this is smaller than
a what means of the complex functions
also we put this some outside right um
so this is also applied to the context
of probability

Theory um and then we can arrive at this
uh yeah this

equations so formally it's called an
upper bounds uh this one right a
quantity that approach from the above
right so this is also means that this is
a free energy um when we find a

posterior that minimize this free energy
 term we will have a chance of
 approaching the lower surprise points u_m
 which potentially increasing the u_h
 modor evidence
 so after we have this terms right here
 we can just u_h put the minus inside and
 then it's become this form this is the
 conventional form that we call variation
 of re energy u_m and after decomposing
 this we will have these two form u_h
 which is the K Divergence of between the
 posterior and the prior u_h minus the
 weighted sum u_h of the lock of of u_m the
 u_h u_m P of or given s which is the
 posterior
 functions or no this is
 likelihood so basically u_m the K
 Divergence term is called the complexity
 and the minus I_p of our given s is
 called accuracy term where the
 complexity terms comput how u_m supplies
 are State u_m to State distributions
 posterior and the prior are far away
 from each other or not right and the
 accuracy terms u_h defines how accurate
 our model is predicting u_m the actual
 observations right so this is also
 relatable to the variational auto
 encoder where u_m we comput the loss of
 the variation Auto encod u_m to be the K
 Divergence of the posterior and the G_{in}
 distribution with a mean of zero and a
 standard deviation of one and then we
 also have the u_h mean Square arrows of
 the actual predictions of the images
 right u_m and then VIP and MF is a little
 bit u_m different than from variation of

out coder VP is variation of free energy
and mfp is marginal free energy as in
the um active inference tutorial um the
paper U so marginal free energy says
that um we compute the CH Divergence
between the poster and the prior and the
prior here is computed from the
transition model um not the um not the
normal distributions of mean of zero and
this standard equation of
one so uh here is the uh General model
of trying to build a deactive inference
model right so here's first of all we're
trying to encode right using the
posterior functions so in active
inference this is the $p(s|o)$ given o is the
transpose of the likelihood Matrix P of
given s this is normally learns with the
cical distributions and using the orally
conjugate
prior um and then uh in the Deep
learning literature um this is often
known as the encoder U that's uh Often
parameterized by the neuron Network um
and then o here is often be an image and
state here is just um some
hidden vectors that we are trying to
compute and then next one is the um La
function right so normally we text in
the states um of um the agent and then
we're trying to estimate the uh
predictions the observation predictions
right um so in traditional active INF
this is known as the likelihood Matrix
or likelihoods um per se um this is also
learns with the Cal categorical
distribution and using the conjugate
prior and deep learning literature um

this is known often known as the decoder
um that is often parameterized by a Ne
Network so uh as we know the variation
of free energy formula the complexity
terms K Divergence here right
and the um uh accuracy term is the minus
l p right given s um so here we compute
the K Divergence between the encode
States and the actual prior States here
and then um next one we compute the um
actual L probability of the predicted
observations and the actual observations
right here
um and that makes the um whole
objectives where we compute the two uh
objectives right here that's uh also be
the um variation of free energy
formula so we just talk about how
inference um inactive instrument work
but there's more we want to build an
inference system that can interact with
the world right so when it comes to
interaction between the agent and and
the environment we also need to have
action
and um also roll out in time step right
so basically we have another elements
here which is at and remember I also
mentioned the transition model right so
after we take an action in the world um
we um the the world um or the
environments evaluates one step right
and it uh it change its own State and
then emits a new observations um and
with we as agents or intelligence system
also try to predict um given the action
in the current states what is the next
States and what and this States is the

prior over u the states of the next
town step where we have not made the
observation O_{t+1} yet so after we
have the S_{t+1} as prior u we do the
same process as I have mentioned before
so u we make up observations u try to
change u our belief u after making
that observations and then learns from
the u new observations to take our
prior into the
posterior so here we learn it u using
the variation of free energy or marginal
free Energy
Formula so as we can see from the
previous slides u we do need somehow U
to somehow plan the actions to interact
with the world so if we're trying to
minimize or surprise u with the
current observations u so in the action
planning we aim to take actions that
minimize the similar free free energy
term in the future u this is often
called that's the expected free energy
by the way u so in here u we introduce
some new terms right Θ u is the
model parameters u the reason why this
is here is because we want to also
compute the u
 u modal the surprise over the modal
parameters or the uncertainty of the
modal parameters u so you can see that
the expected free energy in the future
it's kind of similar to the version of
free Energy Formula it's only needs u
another model of parameters
parameters and the given the π or the
actual plant's policy or plant actions
so after a lot of derivations right I'm

we're not getting into like the real deviation here but um here's the end results that's um something something minus um minus the weighted means of the uh L P of OT right so I'll just focus on this um term is because this is the um goal of the talk today so basically um this term $\log p$ is the expected free energy derivations on of the instrument uh instrument mental term so basically uh this also means that we are trying to maximize the now we're trying to take the action that maximize um the future observations that matches our um prior parap distributions or what we want right so normally um to learn to take actions based um uh to learn to take actions um to align with what we want to see in the future um we of needs to craft and provide the priate preference manually in advance um and then we T the option based um we take the options um that's maximize the likelihoods um given that distributions um but this is often not really practical in the realistic World um H because we cannot like craft the manual distributions like every single time we encounter new environments or encounter a new task right so what we can also do is we can also provide a goal Stat or goal observations directly to the agents which is more practical um but in the end the prior preference distribution is very sparse um and this provides little to no learning signal to the agent as you can see on the figure on

the right and the left right so left is
a prior preference distributions and
right is a single goal distribution State
distributions or goal State
image so uh what can you do to
dynamically uh produce um the uh uh
prior preference right we can try to
learn a prior preference model right
that um produce a goal image at each
time
step so here is on top is the prior
preference model that predictions right
and uh in the bottom here is the model
prediction of the future um given the
sequence of planned actions so we are
trying to say that we um we want to take
the actions um that have uh the roll out
in the Futures predictions matching the
prior preference uh roll out in the
future so how do we do that we first
have to learn the model first
right so in learning uh we can try and
say that each time we achieve a goal
then then we Mark the whole trajectory
as the prior preference trajectory or
the um positive um trajectory or expert
trajectory and then for each
observations marks as go we use a deep
learning like another deep learning
model to estimate start observations um
for example right and then um in
inference time um we actually train when
we train the policy planner uh to
propose the actions right we make the
policy planner proposing the actions
first and then predicts the future
observation based off of that um actions
U that is planned by the policy planner

and then um we actually learn the policy planner based on the signal that is emitted from how aligns the predicted future observation would be um to the predicted goal

um the predict the predicted goal here is the actual prary props model outputs right

um then we train the policy planner with the um mean Square Arrow or um some kind of objectives um that's uh predicts uh uh some kind of objectives between the predicted goal image and the predicted future

observation so there's this problem uh where uh uh it exists between all these learning schemes because it is very computational expensive um so it is generally uh expensive because we have to work directly with the images so for example in learning we have to use a decoder of the um of the PRI prefence model um to actually predict the goal image at each time step right um in inference uh time is also

computationally expensive is because um it has to learn from the images itself and computed down from uh the um the back propagations will have to flow through the decoder back to the action planner so here it comes to the actual methodologies uh we actually use Dynamic PR preference learning um in the state space um instead of the observation space

right so first of all I want to talk a little bit about the world model so uh this uh model architecture is um

basically what I have just um drawn in the background sections but this looks with robots images so in here we have the observations um O_t O_t and here's also the observation the same observations O_T minus one O_T um and we text in the encoder and is the encoder posterior texting the encoder and we have the transation prior um and then we have the decoder right encoder encodes the image in have the states and then decode the um the image the state back to the image and then computes the accuracy loss right accuracy loss uh is minus L prop of the P given S as we have discussed in the variation of pre energy sections and then the complexity is the k k Divergence term between the posterior and the prior right so how do we compute the prior we compute from um the previous steps uh States and then we use the transation model um to estimate the next States prior right so that's um that's how we can get the um prior so uh current work um and popular Works uh regarding the world model or generative models um often inrs um the model called um RSS langage stands for um recurrent State space model um in the translation uh model so it's to zoom out a little bit um we are trying to uh integrate some kind of temporal informations Through Time right um so that's the state of the environment or the states that we're trying to predict be more accurate um and it carries more informations with time um so for example um we cannot um encod the uh enough

information from just a single image
because for example um there can be like
uh position and velocity right the
velocity is acquired um through multiple
images um and cannot be
acquired from a single image if does Mak
sense so what we of to do is we try to
integrate a recurrent NE networks that
text in the previous States um or the
previous recurrent States H_{t-1} um and then we also text in the um
encoded

observations um and then with in
transition um we produce the recurrent
neur network will produce the current
recurrent States um and also predicts
the outputs to to put into the uh
transition model um and also it's the
recurrent stat will also uh be served as
inputs to the posterior Network um so
that's basically uh the
idea and then uh the other way uh around
we could just uh do uh the model as
normal so uh in active inflece asan
texting the actions um that's it belief
uh that would lead to its preferred
outcome uh using instrumental signal um
to construct this prior preference um
Pas has um for example provide a go
statement observations directly to the
agents for example mazaka in 2021 with
contrastive active INF um we can also
manually craft a PR prence distributions
um we can also dynamically produces the
preferred observation that each time set
um but as I mentioned already um all
these method is one way or or another
very expensive um or it's very not

practical or in the reward applications
or um it is very computationally
expensive for dynamically producing the
actual observations um for the the
goals so we can also use Behavior
cloning but it's its trajectory diverges
from on the trading trajectory due to um
small mistakes made by agent as well as
environmental um
stochasticity um and also it is because
of ident IND independent and identically
distributed assumptions of positive
exper data distribution so we can um we
we try we will kind of point it because
it's it's not going to um be uh the same
right the two distributions of the
actual um actual trajectories in the
environment will not be the same as um
every single positive trajectory of the
expert
data so to tackle this we try to
leverage a small quantity of seat
imitation data to learn an uh a neur
network that's dynamically produced
preference Over States not observation
but State um at each time step so this
effectively um produces and easily
generated dense instrumental go signal
um and we also design the agent such as
that it moves according to the
trajectory that is is shaped toward its
own estimation of the future States or
future preferred States so it's trying
to notch the its own trajectory on
toward watchs the imitation positive
data distribution so as you can see on
the figure on the left um so you can see
that in Behavior cloning it's trying uh

the actual trajectory is diverges away from the actual imitation data where the prior preference over the states trying to get uh trying to be shield toward the imitation data itself using the model

signal and r_a is our agent u_m is called robust active inance u_m and it learns u_m from both epistemic signal and instrumental signals but in here we'll just focus on the instrumental signals and how to construct it

so uh yeah so basically we use the decoder u_m or the likelihood functions of the generative model to only visualize on the preferred observations u_m not that in training or test time we do not use the motor learning uh we do not use the actual observation uh goal observations by itself we only use the hidden States so the goal here is to u_m get the states the goal States at each time step be predicted by the prior preference model right u_m You can see that's uh on the top here is the real trajectory and the bottom here is the actual u_m dream that's the dream of the robots that it wants to be right u_m and then we have predicting that dream

right so here is the actual model of the Contra recur and state prior preference learning u_m uh so basically it has u_m it has also similar architecture to the rssm to the world model u_m it also has have an encoder u_m here it also have translation u_m but it does not have the

decoder um so how do we actually uh learn this

uh um prior preference model so first of all uh we uh learn the world generative World model as normal right um but when we say whenever um it's uh whenever the trajectory is uh positive or a successful trajectory um we try to contract um the state of the uh crispy I just call it crispy um we try to contract the state of crispy to the um world model States and then um whenever the um trajectory is negative we try to push away the states um from each other so that the crispy model um can only predict the um positive States so how do we determine um if it is a POS trajectory um or the negative trajectory without having a very sparse signal so we will have a thing called brow or the prior preference signal so how do we compute this raw

signal so imagine um this is the real trajectory um and it is a successful trajectory for example right so we're trying to compute um the prior preference states by having the rate of the successful States is at one and it starts from one and it can be case backward so the case um very fast and um if so in this like really far away States from here um they do not have um enough signal because there's no goal um like there's no direct go from that states and then there's real trajectory that is failed right um You can see that the

mountain car cannot reach um the fls on
the right here so whenever it's built um
we can just PL plat the very very last
observation as minus one and then we
start to De Decay backward
negatively
um and then uh we have the prior
preference rate
signal as um very
negative so the reason why it is it has
to be last is because um we want to um
we want to blame the L observations um
hardest so um in order to uh in order
for the agents to just push away from
the failing States very very very strong
um in the very last stage
so in the real trajectory that have like
multiple successes um we try to put all
the one in all the successful States and
then just Decay
backward um as similar to how we Decay
backward in a single success
case so with that s we now uh know how
to comput um the RO and then this will
serve as the dynamic signal um to the
contrasted loss and it creates a dynamic
behavior of the contrasted loss um
that's trying to either push away or um
contract um the two states of the
preseed state and the world model
States so for example in here um there's
if there's positive row um the
contracted block would be um Contra
Contracting all the states together and
maximizing the similarity between the
precipitate states and the world model
States and then we also learn to
minimize the K diversions between the um

complex uh uh the posterior and the prior uh of over the states of the crispy

model um this is because we want to estimate the translations right so we translation to the um next goal State and then this goal state has to match the actual goal um that is the current stage of the enviro of the rsm model or the world model

um

so in contrast right if row is negative then we're trying to push away um all these states away right so we're trying to maximize this um distance or minimizing the similarity between the um world model States and the cryst spe State and in here notes that there's no complexity objective is because we don't want to um predict the next goal that is um the failing State that's basically the the

idea so that arrives at the uh general formula for the objectives of the contrast State PR preference model um basically uh it is just masking out um The Sign of the row and then multiply with the the K Divergence or the complexity uh between the posterior and the prior of the crisby model um and then we uh minus the row and SE of um the actual predicted State and the world model state so same here can be very much um uh any kind of similarity functions for example um cosine similarity um

and then in the in our work we actually use the actual coine

similar

so in training time we have we have just talked about training time right so we have successfully assuming that we have successfully um learned the prisp model that predicts the future goal um and in inference time we've used this um crispy transation model to estimate future rollout trajectory of the go right so we can see that um on the top here is the observation um actual observations and in the second row is the actual prior preference roll outs or predictions of the future goal uh given the um given the current

observation so given the uh PL actions we also training the plant uh the plant actions right so the actor trying to predict actions given um the observations the predicted observation in the future um so we're trying to say uh this is uh we're trying to compare uh the actual PRI prior preference roll outs with the actual predictions given the actions

um and then we are trying to uh learn the action planner based on the signals or how diver um the future goal is to our future predictions of the world given the

options so uh there can be so many um there can be so many reinforcement learning agents and active in agents that can utilize this technique um of Prior preference learning so active INF agents um I put my works here um it's ra and then there's also Chris bakley's work and baren M work about deep active

influence in value policy gradients
methods and also Auto work is is
the scaling of the act active
influence in PDP and not in car
environments and so on right so
reinforcement learning agents there's
all these sorts of after critics
algorithms like dreamer series soft
after critic dreaming ddpq tqc nd3
and so on so there's a lot of agents
that can utilize the framework because
there's no change in the original
architecture there's just an add-on
to how an agent would apply
the private preference learning signal
into the
system so is the actual flow chart of
of the agents basically we start the
agents and for each episode we evaluate
the environments using the world
model and the policy planner actually
basically we evaluate in the
environment we let the robots play we
let the robots do the task and then we
collect the data and add them to the
replay buffer for example right and then
we can get the main training agent
logic and then first of all we sample
the training data from the buffer so in
here we trying to have we also have
to use The self-imitation Learning
buffer
so yeah so self-imitation learning
is very good technique normally to
balance the number of successful and
failing
trajectories and we also learned so
first of all we learned the world model

encoder decoder transation on rssm and
then we uh train the epistemics model
but in this case we don't need to care
about it um and then next is the um
Contra Rec St PR preference model um
from the data and then um next we
imagine the future trajectory using the
world model and the policy planner right
we try to estimate the future options um
to be um what is the optimized future
actions for example which we estimates
the future action based on how uh
current state
is and then the world model will be
continuing to bouts in the Futures and
then we dream of the futures
and next is the part that we actually um
infer on the uh crispy model we use a
crispy model to imagine the future
prefence
trajectory um given the goal um given
the current observations and we predict
the future goals roll out goals right
so after that we learn the policy
planner right using the instrumental
signal the instrumental signals here is
computed from the um difference um
between the future goal and the um
predicted future dream of the
world right and then with that the
policy planner can learn from uh from
the uh instrumental
signal so here's our um preliminary
results on um active inflece agent so
basically um we can see um the two top
rows is the mountain and car
environments and the two bottom row is
Rob robotics met a world um push button

environment so let's look at the first row for example the observation here is a mountain car have reached the goal and for each um observation here we uh we predict a preference roll out or preference um or a goal State at that single time step so in here we can see that um if the car has not have the momentum yet um then we have to build the momentum by predicting the car going to the left for example right and then after it has built some kind of momentum we trying uh the actual prior preference model predicting the car would go to the right go forward right when it goes here and the prence model will predict the car to be um going closer to the goal something like this so the same applies to the um robot environments where the actual observation is the third uh row and the uh prior preference roll out is on the fourth row so we can see that uh it is also trying to estimates the dream or the dream of the goal of the robots is to actually push the button so um there's different kind of experiment setup that we are trying to learn um basically we have uh four trials on each environment and then we train the experts um using as sof critic or po proximal policy optimization um to collect um a small amount of seed data normally is about uh 20 episodes something um 20 to to 30 episodes um so we test mostly on three mains environments or suits um first of all is Mountain car there's only one task um metal W is 13 task so here's um

all red robots here's metal war and then the robot suit we have two task and then um the C regarding camera setups uh we use um three views for met world and four views for um the robo suit so the green check mark here denotes uh that the robots has uh successfully solve the task um here's a robo suit door open door and then um here's Robo suit lift where it can successfully lift the block and here's more U solving agent tasks so basically agent with Crispy exhibits and ability to perform successfully early on um and uh so agent with without crispy uh struggle to actually succeed on solving the environment um and we also test on another other environment as well for example gasum robotics or and risk task um you can you can see in the middle image on over here so actually you go to the discussion sections so actually um I want to make a note that um with respect to the crispy subm model um RM model utilize contrastive objectives to adapt its parameters so basically it solely learns on um estimating the world model in coted States while pushing itself away from undesired World model States or trajectory so this does not require or the learning of the of the decoder so we we only make sure of the Learned decoder in the generative World model um to visualize the uh to visualize the preferred observations um from the estimated

crispy States so experimentally we remark that this is an um that's we only use um this decoder um to as an auxiliary tools um to prove um that the states of the contusive recurrence Prime preference model uh is useful so theoretically Behavior clotting will be unable to achieve a great uh success in multile multi goal environments due to its Reliance on a fix and finite size uh imitation sample pool um the expert trajectory in this fixed pool May further differ from the actual trajectories um that the agent needs to take to solve the problem at hand um for example there's a distributional gap in in observation and there and um it requires different sets of actions to be taken for example right um so on the other hand right crispy learns to produce dynamically um the goal sticks and not the go observation right um so we can empirically confirm in simulation that our agent um with Dynamic prior preference learning does not suffer from the goal miss much problem uh in the training phase so this is very applicable applicable to the multi-goal environments so um we can see that um this P preference learning uh techniques has a lot of practical applications in different robotic tasks um with for example those with potential dangers in the operations um and functioning for example when human safety must be

considered right um for example a um
trying to drive a autonomous driving car
system where uh it has to have or it has
to predict um the future um goal right
the future goal is not to hit the human
for example right um and we can also
make a lot of advancement in imitation
learning in research community and it
can be um applied by the wide range of
um reinforcement learning in active
inference

agents um and then the last one is to
scale the the active inference agents on
um the real world robotics

agent so future work words actually
considered um importing or integrating
the dynamic PR preference learning in
latent State space model um for example
variation of Dynamics discret variable
Auto encoders or tension waiting models
um and also uh there's another models uh
physical neur robotic system um which
entails an embodied and active and
survival oriented formulations of active
perception and World model

learning so in this uh talk um uh we
have discussed a lot of ideas about
being able to estimate the particular
goal or preferred State um at any
particular time step is very useful um
for cognitive control agents so as our
results demonstrates the agent can then
learn to adapt the tech actions that
leads to um a estimated goal um and then
um in contrast to the approach is taken
in the imitation learning and behavior
cloning literature um which train the
agent's policy based solely based on the

fixed collected expert data sets in our framework the expert signal is estimated dynamically um through an Adaptive prior preference model right closing this Gap the domain gap between the actual trajectories and the collected trading imitation trajectory or preferred trajectory um and then we also discussed the generative model um and and coder decod transations and how the variation of free energy work and then how to learn the model and then we finally we discussed the contrasted recurrent State prior preference model um with a self-revision mechanism and then discuss the model learning and then inference um object deted um and how to integrate stat into the agent planner learning system and then we show the results for um the robotic task and then I mention how it can be applied to different

fields so that concludes my presentation um and thank you very much for um listening thank you that was amazing really great job with the background math done only like a PhD student can but it was great to see how you brought it all together and connected your research to it all at the end so thank you that's all for now see you later thank you see

you

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